

Pain and Analgesia

Pain sensitivity (SCN9A, COMT, FAAH), opioids (OPRM1, CYP2D6, CYP2B6, CYP3A5, ABCB1), anti-inflammatory (CYP2C9), fibromyalgia scores and chronic low back pain, and load of chronic pain loci

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	8
6. Favourable	9
7. Unfavourable	10
8. Mechanisms of ageing	11
9. Practical suggestions	12
10. Limitations	13
11. Methods and sources	14
12. Annex: all variants analysed	15

1. Summary

Favourable. Genetic fibromyalgia is low to typical (16.0 and 34.4 percentiles; loci load at 40.1). The sensitivity to base pain tends to be low: **COMT Val/Val** (G/G; less sensitivity to pain), **SCN9A R1150W** absent and **FAAH C385A** heterozygous (more anandamide, an endogenous analgesic). The **CYP2D6** is *1/*2 probable (normal metabolizer; no number of copies evaluated), so codeine and tramadol are normally activated. **CYP2B6** does not have *6. The multisite chronic pain loci load is typical (55.3).

Unfavourable. Chronic genetic low back pain is at percentile 86.6 (the same score of \$50). The loci load of prolonged use of opioids and analgesics is high (98.4), but it is based on only 3 loci. **CYP2C9** is *1/*3 (intermediate metabolizer): anti-inflammatory drugs (NSAIDs) spend more time in blood, with more gastric and renal risk. The **OPRM1 A118G** is heterozygous and the **COMT Val/Val** also requires a little more opioid for the same effect.

What does it mean? The profile is low sensitivity to acute pain, with a genetic tendency for chronic low back pain (high score), although multisite pain and fibromyalgia are typical or low. The measures with more return are strength and trunk training, sleep and inflammation control, with analgesics chosen by pharmacogenetics. Attention to **LDN 4.5 mg**: it is an opioid antagonist and blocks the analgesia of opioids (codeine, tramadol, morphine, oxycodone, fentanyl). Isotretinoin often causes muscle, joint and lumbar pain; if back pain worsens, it is the first suspicion.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 12 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 3 PGS Catalog scores calculated locally with 3,232,801 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 65 loci, 60 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk (fibromyalgia, chronic low back pain).

Chronic pain

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Fibromyalgia	PGS012561 (Kerrebijn 2026, PRS-CS, 1.11 million variants)	1,081,256 1,113,197 (weighted coverage 97%)	16.0	LESS LIKELY	favourable
Fibromyalgia	PGS012560 (Kerrebijn 2026, PRS-CS, 1.12 million variants)	1,086,330 1,118,760 (weighted coverage 97%)	34.4	TYPICAL	typical
Chronic low back pain	PGS004926 (Tsepilov 2023, SbayesR, 1.09 million variants)	1,065,215 1,090,574 (weighted coverage 97%)	86.6	MORE LIKELY	unfavourable

4. Key variants by topic

Variants of pain sensitivity and pharmacogenetics of analgesics The diplotypes CYP2D6, CYP2C9, CYP2B6 and CYP3A5 are already in the profile; here they are read for pain.

Pain sensitivity

The sodium channel Nav1.7 (SCN9A) triggers the nociceptors; COMT degrades catecholamines that increase pain; FAAH degrades anandamide, an endogenous analgesic.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SCN9A	rs6746030	R1150W	G/G	chip	A (0 copy/s)	favourable	R1150W (Nav1.7): A allele increases the excitability of nociceptors and pain sensitivity (fibromyalgia, low back pain, arthrosis)	Weak
COMT	rs4680	V158M	G/G	chip	A (0 copy/s)	favourable	Val158Met: Allele A (Met) low to COMT, increases pain sensitivity and reduces the need for morphine; Val/Val (G/G) : less sensitivity to pain and a slightly higher need for opioid	Moderate
FAAH	rs324420	P129T	C/A	chip	A (1 copy/s)	favourable	C385A (Pro129Thr): the allele A low to FAAH (more anandamide, an analgesic endocannabinoid)	Weak

Opioids: target and metabolism

The μ (OPRM1) receptor is the target of morphine and other opioids. CYP2D6 activates codeine and tramadol; CYP2B6 metabolizes methadone and ketamine; CYP3A4/5 metabolizes fentanyl and oxycodone; P-gp (ABCB1) removes opioids from the brain.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
OPRM1	rs1799971	A118G	A/G	chip	G (1 copy/s)	without judgment	A118G (Asn40Asp): G allele reduces μ receptor expression; it asks a little more opioid for the same effect. Also modulates the response to naltrexone	Moderate
CYP2D6	rs3892097	*4	C/C	chip	T (0 copy/s)	favourable	*4 (zero allele): without activation of codeine and tramadol in morphine/O-demethyltramadol	Strong
CYP2D6	rs1065852	*10	G/G	chip	A (0 copy/s)	favourable	*10 (reduced function)	Moderate
CYP2D6	rs16947	*2	A/G	chip	A (1 copy/s)	neutral	2 (R296C): normal function; with the SNPs above defines CYP2D6 1/*2 probable (normal metabolizer; no copy number)	Moderate
CYP2B6	rs3745274	516G>T	G/G	chip	T (0 copy/s)	favourable	*6 (516G>T): less CYP2B6 (methadone and ketamine more exposed)	Strong
CYP3A5	rs776746	*3	C/T	chip	C (1 copy/s)	neutral	3: loss of expression; 1 (T) = expressor. Fentanyl, oxycodone and alfentanil are mainly CYP3A4; the clinical effect of CYP3A5 is small	Moderate
ABCB1	rs1045642	C3435T	A/A	chip	A (2 copy/s)	without judgment	C3435T (A in strand + = T): P-gp transporter in the blood brain barrier; inconsistent association with dose and adverse effects of opioids	Weak

Anti-inflammatory (NSAIDs)

CYP2C9 eliminates most NSAIDs. With reduced function, they spend more time in the blood, with a higher risk of ulcer, hemorrhage and kidney injury.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2C9	rs1057910	*3	A/C	chip	C (1 copy/s)	unfavourable	*3 (Ile359Leu): very reduced function; NSAIDs (ibuprofen, celecoxib, meloxicam, piroxicam) with more exposure and gastrointestinal/renal risk	Strong
CYP2C9	rs1799853	*2	C/C	chip	T (0 copy/s)	favourable	*2 (Arg144Cys): reduced function	Strong

5. GWAS loci burden

Sum of the loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of chronic multisite pain and general factor of pain, fibromyalgia and prolonged use of opioids or analgesics. Alleles are oriented towards more risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Multisite chronic pain	45 (42)	37 of 84	42	55.3	typical
Fibromyalgia	17 (15)	14 of 30	15	40.1	typical
Prolonged use of analgesics and opioids	3 (3)	5 of 6	3	98.4	unfavourable

6. Favourable

- **Fibromyalgia** in the percentiles 16.0 and 34.4 (loci load at 40.1). **Moderate.**
- **COMT Val158Met G/G (Val/Val):** Active COMT; less sensitivity to pain and less risk of chronic generalized pain in COMT studies. **Moderate.**
- **SCN9A R1150W G/G:** without the Nav1.7 allele that increases pain. **Weak.**
- **FAAH C385A C/A:** more anandamide (endogenous analgesia and less anxiety). **Weak.**
- **Probable CYP2D6 *1/*2 (normal):** codeine and tramadol with normal activation (without number of copies evaluated). Strong for alleles tested.
- **CYP2B6 without *6** (*2/*4 in the profile, normal to increased activity): methadone and ketamine are no longer exposed (may even get a little lower levels). **Moderate.**
- **Multisite chronic pain:** typical loci load (55.3; 45 loci). **Moderate.**

7. Unfavourable

- **Chronic low back pain** at percentile 86.6 (the same as \$50). **Moderate.**
- **Prolonged use of opioids/analgesics:** loci load at percentile 98.4 (only 3 loci; association mainly linked to chronic pain and dependence). **Weak.**
- **CYP2C9 *1/*3 (intermediate metabolizer):** ibuprofen, celecoxib, flurbiprofen, meloxicam and piroxicam with more exposure. By pharmacogenetic guidelines: starting with the minimum dose, meloxicam to 50% and avoiding piroxicam and tenoxicam (very long semicircle). **Strong.**
- **OPRM1 A118G A/G:** less μ receptor; in surgery, it asks for a little more morphine. **Weak moderate.**
- **COMT Val/Val:** on the other hand, the need for morphine tends to be slightly higher than in Met/Met. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Neuroinflammation	Chronic pain involves central sensitization and glial activation; load of typical multisite pain loci (55.3)	Neutral
Chronic Inflammation	Low back pain and arthrosis are maintained by low-grade inflammation; low back pain at percentile 86.6	Unfavourable
Remodelling of the extracellular matrix	Degeneration of the intervertebral disc and cartilage in chronic low back pain	Unfavourable (mild)
Intercellular Communication	FAAH C385A: more endocannabinoid signaling; COMT Val/Val: less pro-pain catecholamines	Favourable
Sleep disturbance	Pain and sleep reinforce each other; sleeping poorly the threshold of pain in the following day	Conditional
Fragility	Chronic back pain reduces physical activity and accelerates loss of strength with age	Unfavourable (conditional)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Force training with trunk and gluteal work** 2-3×/week (e.g. hip fold, boards, bird-dog, bridge) and daily walking. It is the treatment with better evidence for chronic low back pain (see also §50). *Suggestion (not applied).*
2. **If you need a NSAID (CYP2C9 *1/*3):** the lowest effective dose during the shortest time, preferably ibuprofen or short-term naproxen. Meloxicam to 50% of the dose and avoid piroxicam and tenoxicam. For localized pain, prefer diclofenac in gel. *Suggestion (not applied).*
3. **NSAIDs with telmisartan and dapagliflozin:** the combination increases the risk of acute kidney injury, especially with dehydration. Use NSAIDs only for a few days, well hydrated, and knowing creatinine. *Suggestion (not applied).*
4. **Paracetamol with elevated ALT:** not to pass 2-3 g/day while the ALT is high, and never with alcohol (already zero). *Suggestion (not applied).*
5. **Opioids and LDN 4.5 mg:** LDN blocks opioids. If an opioid is needed (acute pain or surgery), suspend LDN 24-72 hours before, if planned, and inform the doctor. Codeine and tramadol work normally (normal CYP2D6). Avoid prolonged use of opioids. *Suggestion (not applied).*
6. **Register in the clinical process:** CYP2C9 *1/*3, CYP2D6 *1/*2, CYP2B6 *2/*4, OPRM1 A/G and LDN. *Suggestion (not applied).*
7. **7-9 hours sleep** (melatonin 0.3-1 mg of the plan; caffeine ≤ 200 mg only in the morning): short sleep increases the pain of the following day. *Suggestion (not applied).*
8. **Isotretinoin:** if they arise or worsen back pain, muscle or joint pain, discuss with the doctor the dose or a pause before looking for another cause. *Suggestion (not applied).*
9. **Omega-3 3.2-4 g/day (plan) and vitamin D of the plan:** modest anti-inflammatory effect on joint pain. For episodes of pain, local heat, physiotherapy and techniques of gradual exposure to movement. *Suggestion (not applied).*

10. Limitations

- CYP2D6 was read by SNPs; deletions and duplications of the gene (copy number) were not evaluated, and an ultrafast metabolizer cannot be excluded.
- The effects of COMT, OPRM1, SCN9A and FAAH on pain are small and vary between studies.
- The score of chronic low back pain is the same as \$50; it is not a new and independent sign.
- The scores of fibromyalgia (Kerrebijn 2026) were trained in health records.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (3 scores: two of fibromyalgia and one of chronic low back pain). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** International pharmacogenetic guidelines for CYP2C9 and NSAIDs (Theken 2020) and for CYP2D6, OPRM1 and COMT with opioids (Crews 2021); Zubieta 2003 and Diatchenko 2005 for COMT; Reimann 2010 for SCN9A; Johnston 2019 for multisite pain; Kerrebijn 2026 and Tsepilov 2023 for scores. GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Pain sensitivity	SCN9A	rs6746030	G/G	chip	A	0	strand +	without the risk allele
Pain sensitivity	COMT	rs4680	G/G	chip	A	0	strand +	without the risk allele
Pain sensitivity	FAAH	rs324420	C/A	chip	A	1	strand +	1 copy of the protective allele
Opioids: target and metabolism	OPRM1	rs1799971	A/G	chip	G	1	strand +	1 copy of effect allele
Opioids: target and metabolism	CYP2D6	rs3892097	C/C	chip	T	0	strand +	without *4: normal activation of codeine and tramadol
Opioids: target and metabolism	CYP2D6	rs1065852	G/G	chip	A	0	strand +	without *10
Opioids: target and metabolism	CYP2D6	rs16947	A/G	chip	A	1	strand +	Probable 1/2: normal metabolizer
Opioids: target and metabolism	CYP2B6	rs3745274	G/G	chip	T	0	strand +	without *6: methadone and ketamine without increased exposure
Opioids: target and metabolism	CYP3A5	rs776746	C/T	chip	C	1	strand +	1/3: expression (less common in the reference population)
Opioids: target and metabolism	ABCB1	rs1045642	A/A	chip	A	2	strand +	2 copy of the effect allele
Anti-inflammatory (NSAIDs)	CYP2C9	rs1057910	A/C	chip	C	1	strand +	1/3: intermediate metabolizer (NSAIDs: minimum initial dose; avoid those with long half-life)
Anti-inflammatory (NSAIDs)	CYP2C9	rs1799853	C/C	chip	T	0	strand +	without the risk allele